

Xi-Zhu Wang

ASIC / Digital IC Design & Verification

z790305000@gmail.com | +886 972-893-532 | Hsinchu, Taiwan
github.com/xizhuwang | xizhuwang.github.io | Available Sep. 2026

Summary

M.S. in Electrical Engineering with hands-on experience in RTL design and verification, synthesis, and physical implementation, plus training in SystemVerilog assertions and formal property verification. Built pipelined processors and CPU-accelerator systems, and researched mixed-precision HBM-PIM for multi-user LLM inference. Seeking digital IC design or verification roles in CPU, SoC, interconnect, and accelerator development.

Education

National Taiwan University of Science and Technology (NTUST) *M.S., Electrical Engineering* Aug. 2026
Research: HBM-PIM architecture, mixed-precision computing, and multi-user LLM inference scheduling
National Taiwan University of Science and Technology (NTUST) *B.S., Electrical Engineering* Jun. 2024
Double Major in Finance; senior project on fault-tolerant flash-memory design

Technical Skills

RTL / Verification	Verilog, VCS, Verdi, ModelSim, testbench/regression development; SystemVerilog and SVA (training)
Architecture	Five-stage pipeline, hazards/forwarding, CPU-accelerator integration, HBM-PIM, CNN/LLM accelerators
Implementation	Design Compiler, ICC2, Innovus, Calibre DRC/LVS, timing/area/power analysis, OpenROAD
Programming / FPGA	C/C++, Python, Bash, Linux, Git, Vivado, Vitis HLS, ZCU102, PYNQ-Z2

Selected Projects

Mixed-Precision HBM-PIM for Multi-User LLM Inference 2024–2026

- Designed mixed-precision PIM compute architecture and adaptive scheduling under bandwidth and memory-pressure constraints; evaluated latency, throughput, energy, and accuracy trade-offs with Ramulator2.
- Implemented and verified the compute-unit RTL, then completed synthesis and physical implementation using a TSMC 16 nm ADFP academic flow.
- Co-authored *Adaptive Multi-User Scheduling for Large Language Model Inference on HBM-Based Processing-in-Memory Accelerators*, ACM SAC 2026; DOI: 10.1145/3748522.3779794.

RISC CPU with MNIST CNN Accelerator Apr.–Jun. 2025

- Integrated a CNN accelerator with a 32-bit RISC CPU for end-to-end inference; developed RTL, verification scripts, and CPU-accelerator control/data movement.
- Completed post-layout implementation in a TSMC 16 nm ADFP academic flow; achieved 89% accuracy at approximately 100 MHz, with estimated 2.46 mW dynamic power and 13,440 μm^2 area.

Five-Stage Pipelined MIPS CPU 2023

- Implemented a 32-bit processor with forwarding, hazard detection, load-use stalls, and branch control; verified functionality with ModelSim and exercised an OpenROAD implementation flow.

Timing Correlator and 16-QAM MIMO Detector Sep.–Dec. 2025

- Implemented the Timing Correlator and K-Best-based 16-QAM MIMO detector in Verilog and completed logic synthesis using the 16 nm ADFP academic flow; post-synthesis cell area: 17,655 μm^2 .

Standard-Cell Placement Legalization Jul.–Dec. 2024

- Implemented a C++ legalization engine for approximately 50k cells, emphasizing constraint handling, stable data structures, and reproducible evaluation; reduced average displacement by 18% against the project baseline.

Experience

Teaching Assistant, Computer Organization NTUST Spring 2025
Guided CPU datapath/control, RTL simulation, waveform debugging, and OpenROAD labs; developed Python/Bash grading automation.

EDA Infrastructure Assistant NTUST 2024–2025
Supported shared EDA environments, reproducible regression flows, and synthesis/P&R tool setup for student and research projects.

Selected Training

TSRI: SystemVerilog; Cadence JasperGold Formal Property Verification; Design Compiler Logic Synthesis; ICC2 Physical Design and Verification; Catapult HLS for AI Hardware Architecture.